

Internship Report

Submitted By

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1 Declaration Form

I, **Faiza Munir** Registration No. **FA23-BSE-030**, enrolled in the **BSE (Bachelor of Science in Software Engineering)**, hereby declare that I have successfully completed a **Virtual internship of 12 weeks** at **Masihai PVT LTD**

This report has been prepared based on the tasks and experiences gained during the internship period. The internship was carried out under the supervision of **Shayan Ahmed Mushtaq** at the host organization and under the faculty guidance of **Mr. Waqas Ali** at CUI.

I affirm that this report is an original work and has not been submitted elsewhere for academic credit.

Student: _____

Signature: _____

Date: _____

2 Acknowledgement

I would like to express my sincere gratitude to Masihai Pvt. Ltd. for providing me with the opportunity to complete my internship. I am especially thankful to Mr. Shayan Ahmed Mushtaq (CTO, Masihai Pvt. Ltd.) for his continuous guidance, support, and timely feedback throughout this period.

I also appreciate my seniors and team members who were always ready to help me when I faced technical challenges. Finally, I extend my thanks to my instructors at COMSATS University Islamabad and my family for their encouragement and motivation during this journey.

3 Executive summary

This report highlights my three-month remote internship at Masihai Pvt. Ltd., a technology-driven company working on digital healthcare solutions. My role was focused on app development using Flutter, Dart, Firebase, and App-write, where I contributed to both backend logic and frontend UI/UX.

During the internship, I worked on diverse tasks such as fixing navigation issues, restructuring large code files, implementing state management with Cubit/Bloc, database fetching and updates, and UI problem resolution. I collaborated with my supervisor and team via Google Meet, WhatsApp, and GitHub, ensuring timely completion of tasks.

This internship allowed me to apply concepts from my academic courses, enhance my problem-solving and coding practices, and gain valuable professional skills. Overall, it was a transformative experience that strengthened my career path as a Full Stack Software Engineer.

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Part No 1

4 A brief introduction of the organization and internship

Masihai Pvt. Ltd. is a healthcare technology company that connects providers and patients through an innovative digital platform. Its mission is to **revolutionize healthcare accessibility** by offering digital consultations, patient management, and appointment systems. I worked as a **remote intern** in the **App Development team**, where I was responsible for building and improving mobile application features. My learning objectives included strengthening my skills in **Flutter and Dart**, gaining experience with **Firebase and App-write**, and enhancing my understanding of **full stack mobile development**.

5 Internship Details

5.1 Duration and Timeline

- **Start Date:** 1st June 2023
- **End Date:** 31st August 2023
- **Mode:** Remote
- **Schedule:** Flexible, with task deadlines set according to complexity
- **Total Duration:** 3 months

5.2 Nature of Work

My internship role was focused on **app development**, covering both frontend and backend aspects. I worked on coding, debugging, UI/UX adjustments, state management, and backend logic integration.

5.3 Team Interaction & Reporting

We used **WhatsApp** for daily communication, **Google Meet** for task assignments and progress discussions, and **GitHub** for code submission and collaboration. I could take guidance from seniors in lead roles.

5.4 Tools & Technologies Used

- **Programming Language:** Dart
- **Framework:** Flutter
- **Backend & Database:** Firebase, Appwrite
- **Version Control:** Git/GitHub
- **IDE:** Android Studio

Part No 2

6 Tasks Performed & Project Summaries

6.1 Introduction

During my internship at Masihai PVT LTD, I worked as a Mobile App Development Intern with a focus on Flutter, Firebase, and Appwrite. The internship allowed me to contribute to the development of a healthcare management system while strengthening my technical and collaborative skills. My tasks ranged from UI development and bug fixing to backend integration, database optimization, and restructuring of the project architecture. The following sections highlight the major duties and assignments I performed, focusing on three key areas:

building UI flows and forms, backend integration and database optimization, and project restructuring and performance improvements.

6.2 UI Related Tasks

6.2.1 Objective & Background:

A major part of the healthcare app involved enabling patients to book appointments, manage prescriptions, and interact with doctors through clean and functional user interfaces. The objective was to design responsive forms, doctor availability lists, and reusable UI components that aligned with the project's design system.

6.2.2 My Role & Responsibilities

- Built appointment booking forms and doctor availability lists using Cubit for state management.
- Designed and implemented reusable UI components (buttons, loaders, text fields, typography styles) to maintain consistency across the app.
- Connected navigation between key screens such as appointments, doctor profiles, and settings.
- Performed minor bug fixes in UI layouts (alignment, scrolling issues, formatting errors).

6.2.3 Process, Methodology, and Tools:

- Used **Flutter** and **Dart** for cross-platform UI development.
- Applied **Cubit (Bloc package)** for predictable state management.
- Followed **feature-based architecture** (separating code into screens, cubits, and repositories).
- Debugged and tested UI flows through **emulators**.

6.2.4 Final Outcome/Deliverable:

The outcome was a set of fully functional and reusable UI flows for appointment booking and doctor listings, with smooth navigation and improved user experience. This provided a foundation for integrating backend services and reduced redundancy in UI development.

6.3 Backend Integration and Database Optimization

6.3.1 Objective & Background:

The system initially relied on Firebase, but later migrated to **Appwrite** for greater control and flexibility. The goal was to optimize the database structure, integrate doctor and user management screens with the backend, and ensure smooth data flow without redundancy.

6.3.2 My Role & Responsibilities:

- Helped design a cleaner **Entity Relationship Diagram (ERD)** for the database, minimizing redundant fields.
- Integrated UI components with Appwrite backend for read-only operations, such as fetching doctors' lists and availability data.
- Implemented **pagination** in doctor listings to improve load time and performance.
- Added caching for frequently accessed data, such as doctor availability, to minimize repeated requests.

6.3.3 Process, Methodology, and Tools:

Collaborated with the team during **architecture meetings** to understand migration requirements. Used **Appwrite APIs** to connect Flutter screens with backend services. Applied simple caching techniques and route-aware triggers to refresh data only when needed. Tested integration flows to ensure consistency between patient, doctor, and admin apps.

6.3.4 Final Outcome/Deliverable:

The integration and optimization work resulted in a smoother and more reliable data flow, with improved performance for frequently accessed features. The optimized ERD also laid the groundwork for better scalability and maintainability of the system.

6.4 Project Restructuring and Performance Improvements

6.4.1 Objective & Background:

As the system grew in complexity, maintaining clean code and consistent architecture became essential. The project required restructuring of folders, refactoring of state management, and performance optimization of key modules like Appointment History and Prescription History.

6.4.2 My Role & Responsibilities:

- Restructured project files into a **standardized feature-based layout** (features/appointments, features/prescriptions, etc.).
- Refactored data calls into **repository classes** to prevent direct API calls from UI code.
- Reviewed Cubit/state patterns across features, ensuring consistency with one Cubit per feature.
- Identified and fixed performance issues such as unnecessary widget rebuilds in doctor lists.
- Added UI improvements like **loading indicators** and **basic validation** in forms for better user experience.

6.4.3 Process, Methodology, and Tools:

- Followed best practices in **Flutter architecture** (repository pattern, separation of concerns).
- Used **Git version control** for code updates and commits, ensuring handover was reflected continuously.
- Conducted **integration tests** across different modules to catch synchronization issues.
- Applied **RouteAware mixins** for targeted screen refreshes, reducing unnecessary API calls.

6.4.4 Final Outcome/Deliverable:

The restructuring led to a cleaner and maintainable codebase with consistent patterns across modules. Performance enhancements such as optimized widget rebuilds and caching significantly improved app responsiveness. The final weeks of my internship ensured that the refactored project compiled cleanly, passed integration tests, and was ready for future development.

Part No 3:

7 Learning Experiences

7.1 Knowledge Acquired

During my internship at Masihai PVT LTD, I gained both theoretical and practical knowledge that closely connected with what I had studied at COMSATS University Islamabad (CUI). On the theoretical side, I applied concepts from courses such as **Mobile Application Development, Database Systems, and Software Engineering**. For example, my understanding of **Entity Relationship Diagrams (ERDs)** from the Database course helped me in optimizing the backend structure to minimize redundant fields. Similarly, the principles of **software architecture** and **design patterns** introduced in Software Engineering were practically applied when I worked with feature-based architecture, repository patterns, and state management using Cubit.

On the practical side, I deepened my technical knowledge of **Flutter and Dart** for building cross-platform mobile applications. I learned how to structure projects into maintainable modules, integrate APIs, and apply performance optimization techniques such as caching and widget rebuild control. Working with **Firestore** initially and then migrating to **Appwrite** exposed me to real-world database integration, authentication, and data retrieval processes. This practical exposure bridged the gap between classroom learning and industry practices, making my academic knowledge more concrete and applicable.

7.2 Skills Learned

Throughout the internship, I refined both my technical skills and interpersonal skills.

From a technical perspective, I strengthened my skills in **Flutter development, state management, backend integration, and database design**. I learned how to implement pagination for large datasets, optimize UI rendering, and conduct integration testing across multiple app modules. These skills built upon what I had learned in my **Programming and Database courses** but extended them into a professional environment where efficiency, maintainability, and teamwork was critical.

Equally important were the interpersonal skills I developed. Since the internship was remote, **clear and proactive communication** was essential. I learned to provide progress updates regularly, seek clarification when needed, and collaborate effectively with my mentor. I also improved my **time management** by balancing weekly tasks and ensuring timely commits. Furthermore, working in a team environment sharpened my ability to accept feedback and adapt to changing requirements, skills that were first introduced through group projects at CUI but became far more structured and professional in this setting.

7.3 Observed Attitudes and Gained Values

One of the most valuable aspects of the internship was observing and adopting professional attitudes and values that are critical for a successful career. I learned the importance of **professionalism and integrity**, especially in consistently meeting deadlines and delivering clean, maintainable code. I also recognized the value of **punctuality and accountability** by ensuring my code handovers were reflected in the repository regularly, so that the team could always track progress.

Additionally, I observed a culture of **collaboration and problem-solving**. Team members supported each other in debugging and restructuring tasks, which reinforced the importance of knowledge-sharing in software development. I also learned to adopt a **growth mindset**, where challenges were viewed as opportunities to improve skills rather than obstacles. These professional values are essential not only in the software industry but also for long-term career growth.

7.4 The Most Challenging Task Performed

The most challenging task during my internship was the **restructuring of the Appointment History and Prescription History modules** into the new feature-based architecture. This task was complex because it required moving multiple files, updating imports, ensuring compatibility across cubits, repositories, and screens, and performing end-to-end testing afterward.

To approach this task, I first planned the restructuring in smaller steps, beginning with moving files to their respective folders and gradually updating the import paths. After that, I reviewed the Cubit classes and repository calls to ensure they aligned with the new structure. One obstacle I faced was dealing with compilation errors caused by mismatched imports and model references. To overcome this, I carefully used version control to test changes incrementally and rolled back if needed. I also collaborated with my mentor to clarify how certain modules should be structured.

In the end, the modules compiled successfully, passed integration testing, and became consistent with the overall architecture. Completing this task not only gave me confidence in handling complex refactoring but also taught me patience, attention to detail, and the value of breaking down large problems into manageable steps.

8 Challenges Faced & Solutions

During my internship at Masihai PVT LTD, I encountered several professional, technical, and interpersonal challenges that tested my adaptability and problem-solving skills. One of the major challenges was **adapting to remote collaboration**. Since the internship was fully virtual, I could not rely on face-to-face discussions. Miscommunication sometimes occurred when requirements were not immediately clear. To address this, I made a habit of asking clarifying questions during team discussions and providing regular progress updates through commits and written summaries. This proactive communication not only reduced misunderstandings but also strengthened my ability to work independently while staying aligned with the team.

Another challenge was **applying academic knowledge to real-world problems**. While I had studied concepts like database normalization, ERDs, and software architecture in my courses at COMSATS University Islamabad, translating that theoretical knowledge into practical solutions required effort. For example, restructuring the appointment and prescription modules demanded more than just understanding the repository pattern—it required balancing performance optimization with readability. I overcame this challenge by breaking tasks into smaller steps, reviewing relevant documentation, and seeking guidance from my mentor when needed.

Managing workload and deadlines was also a challenge. Each week came with a set of tasks, and sometimes unexpected bugs appeared that consumed more time than planned. At first, this created stress as I tried to balance fixing issues while continuing with new tasks. I learned to prioritize effectively by focusing on critical bugs first and then tackling lower-priority features. Over time, I became more confident in managing my time, which made my workflow smoother. Lastly, I faced **technical challenges in performance optimization**, such as identifying unnecessary widget rebuilds and ensuring efficient API usage. Initially, these issues were difficult to debug because the symptoms were subtle. By using debugging tools and carefully analyzing rebuild logs, I was able to pinpoint the problems and implement solutions like caching and RouteAware refresh logic. These solutions not only resolved the issues but also enhanced my ability to troubleshoot systematically.

Through these challenges, I developed resilience, better problem-solving strategies, and stronger communication habits, all of which will help me in my professional journey.

9 Reflection & Conclusion

Reflecting on my internship experience, I can confidently say that it was an invaluable step in both my academic and professional journey. The internship gave me the opportunity to contribute meaningfully to a real-world healthcare management system while applying and extending the skills I learned at COMSATS University Islamabad. I was able to connect theoretical knowledge from courses like Database Systems, Software Engineering, and Mobile Application Development with practical applications such as designing ERDs, implementing repository patterns, and building scalable Flutter modules.

This experience has significantly influenced my career goals. I entered the internship with general programming knowledge, but by the end, I developed a stronger interest in **mobile app development and backend integration**, particularly in healthcare technology. Working on modules such as appointment booking and prescription history helped me see how software directly improves user experience, which reinforced my desire to pursue a career in building impactful applications.

On a personal level, the internship helped me grow in confidence, patience, and responsibility. I

learned how to manage my time more effectively, collaborate remotely with a professional team, and adapt to changing requirements. I also developed an appreciation for professional values such as punctuality, integrity, and continuous improvement, which I will carry forward into my career.

The experience has prepared me well for future challenges in the software industry. It gave me the technical grounding in Flutter, Firebase, and Appwrite while also improving my soft skills in communication, teamwork, and problem-solving. Looking ahead, I aim to strengthen my expertise in backend systems, cloud technologies, and advanced state management techniques to become a more versatile developer.

In conclusion, my internship was not just about completing weekly tasks but about developing the mindset and skill set required to succeed in a professional environment. It clarified my career path, enhanced my abilities, and provided a solid foundation for my future growth as a software engineer.

10 Appendices

Attached

- Offer/Appointment Letter from the organization
- Internship Completion Certificate

